Sardar Patel Institute of Technology

Bhavan's Campus, Munshi Nagar, Andheri (W), Mumbai: 400058, India
(Autonomous College of Affiliated to University of Mumbai)

End Semester Examination
December-2024

Max. Marks: 100 **Duration: 3 Hrs**

Class: TE **Semester: V**

Course Code: CS304 **Programme: CE**

Name of the Course: Distributed Computing

Instructions:

1. **All questions are compulsory.**
2. **Draw neat diagrams and write equations wherever necessary.**
3. **Assume suitable data if necessary**

Course Outcomes: At the end of the course students will be able to:

1. **Understand the Principles and desired Properties of distributed Systems.**
2. **Apply the Various Communication techniques for distributed communication.**
3. **Apply the concepts of process, naming, consistency, replication, and fault tolerance in a distributed environment.**
4. **Apply the algorithms such as clock synchronization, election, and mutual exclusion in distributed applications.**
5. **Identify the challenges in developing distributed applications.**

Q. No.		Max. Marks	CO BL
1 (A)	What are the different organizations of processors and memories in Distributed Computer Systems? Describe with the help of a diagram.	10	1-2
1 (B)	What is an open distributed system? How does an open distributed system separate policy from mechanism? Give an example of an Open Distributed System.	10	1-2
2(A)	Describe Blocking and non-blocking types of IPC, Which is easier to implement and why? Discuss their relative advantages and disadvantages.	10	2-2
2 (B)	Why transient synchronous communication has inherent scalability problems, and how these could be solved.	10	2-3
3 (A)	What are threads in distributed systems? What are the different server Implementation schemes? Describe their characteristics.	10	3-2

3 (B)	With the help of a neat diagram explain the principle of maintaining a migration stack to support the migration of an execution segment in a heterogeneous environment. i. What are the three types of processes to resource binding? ii. What are the three types of resource-to-machine binding? iii. What are actions to be taken with respect to the references to local resources when migrating code to another machine	10	3-3
4 (A)	Describe Maekawa's Algorithm for distributed mutual exclusion, give an example with voting set n=4, and highlight its advantages and disadvantages over other non-token-based algorithms.	10	4-2
4 (B)	Suppose that two processes detect the demise of the coordinator simultaneously and both decide to hold an election using the bully algorithm. What happens? Explain the working of Bully Algorithm with Example. OR How Mutual Exclusion is handled in Distributed System? Compare the Centralized, Distributed and Token Ring Algorithms with respect to Message per entry/exit, Delay before entry (in message times), and Problems.	10	4-2
5 (A)	During the discussion of the consistency model, we often referred to the contract between software and data store. Why is such a contract needed?	10	5-3
5 (B)	Suppose you have decided to use the high-low policy as the process transfer policy of a load balancing algorithm for distributed system. Select the suitable method that you will use in your implementation for choosing high mark and low mark values. Do these threshold values have to be same for all processors in the system? Give reasons for your answer.	10	5-5